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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

PROJECT TITLE

Hospital Patient Registration and Appointment Scheduling System

Submitted by

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Semester: **II**

DECLARATION

I hereby declare that the project entitled **"Hospital Patient Registration and Appointment Scheduling System"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Mr. Dharaneesh** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Menaka C.M

CERTIFICATE

This is to certify that the project entitled **"Hospital Patient Registration and Appointment Scheduling System"** is a **bona fide work** carried out by **MenakaC.M, [1-'C'/2025-2026], B.Sc.Computer Science KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Javaprogramming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: 10-04-2026

Submitted for the Viva-Voce Examination held on. 10 /04/2026

Guide:
Mr Dharaneesh.

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Head of the Department
Dr.Boobalan

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ABSTRACT

The Hospital Patient Registration and Appointment Scheduling System is a C-based application designed to simplify hospital management processes. In many hospitals, patient registration and appointment booking are done manually, leading to errors, delays, and inefficiencies.

This system automates patient registration, stores patient details, and schedules appointments efficiently. It helps in reducing waiting time, avoiding appointment clashes, and maintaining accurate records. The system uses core C programming concepts such as structures, file handling, loops, and functions to manage data effectively.

1. INTRODUCTION

Hospitals require an efficient system to manage patient information and appointments. Manual systems are time-consuming and prone to errors.

This project provides a computerized solution that:

- Registers patients digitally
- Schedules appointments
- Maintains records efficiently

The system improves hospital workflow and reduces administrative burden.

1.1 Objective

- To develop a patient registration system using C
- To automate appointment scheduling
- To reduce manual errors
- To maintain patient records efficiently
- To improve hospital management

1.2 Overview

The system performs the following functions:

- Patient registration
- Appointment booking
- Viewing patient records

2. PROGRAM ANALYSIS

2.1 Problem Definition

The existing manual system has several drawbacks:

- Time-consuming registration process
- Appointment conflicts
- Data redundancy and errors
- Difficulty in maintaining records
- Risk of data loss

2.3 Scopes and Limitations

Scope

- Used in small hospitals and clinics
- Helps manage patient records
- Improves scheduling efficiency

Limitations

- No online booking system
- Limited to basic functionality
- Requires manual data entry

3. SYSTEM DESIGN

The system is divided into modules for better functionality and management.

3.1 MODULES

1. Patient Registration Module

Collects patient details (Name, Age, Gender, Contact)

Stores data in file

2. Appointment Scheduling Module

Assigns appointment date and time

Prevents scheduling conflicts

3. Record Management Module

Displays patient records

Searches patient details

4. File Handling Module

Stores and retrieves data

Maintains permanent records

4. METHODOLOGY

4.1 PROBLEM ANALYSIS

- Identified issues in manual systems
- Need for automation

4.2 SYSTEM DESIGN

- Divided system into modules
- Designed workflow

4.3 PROGRAM DEVELOPMENT

Used:

- Structures (struct)
- Functions
- File handling
- Loops and conditions

4.4 TESTING AND VALIDATION

- Checked data storage
- Verified appointment scheduling
- Fixed errors
-

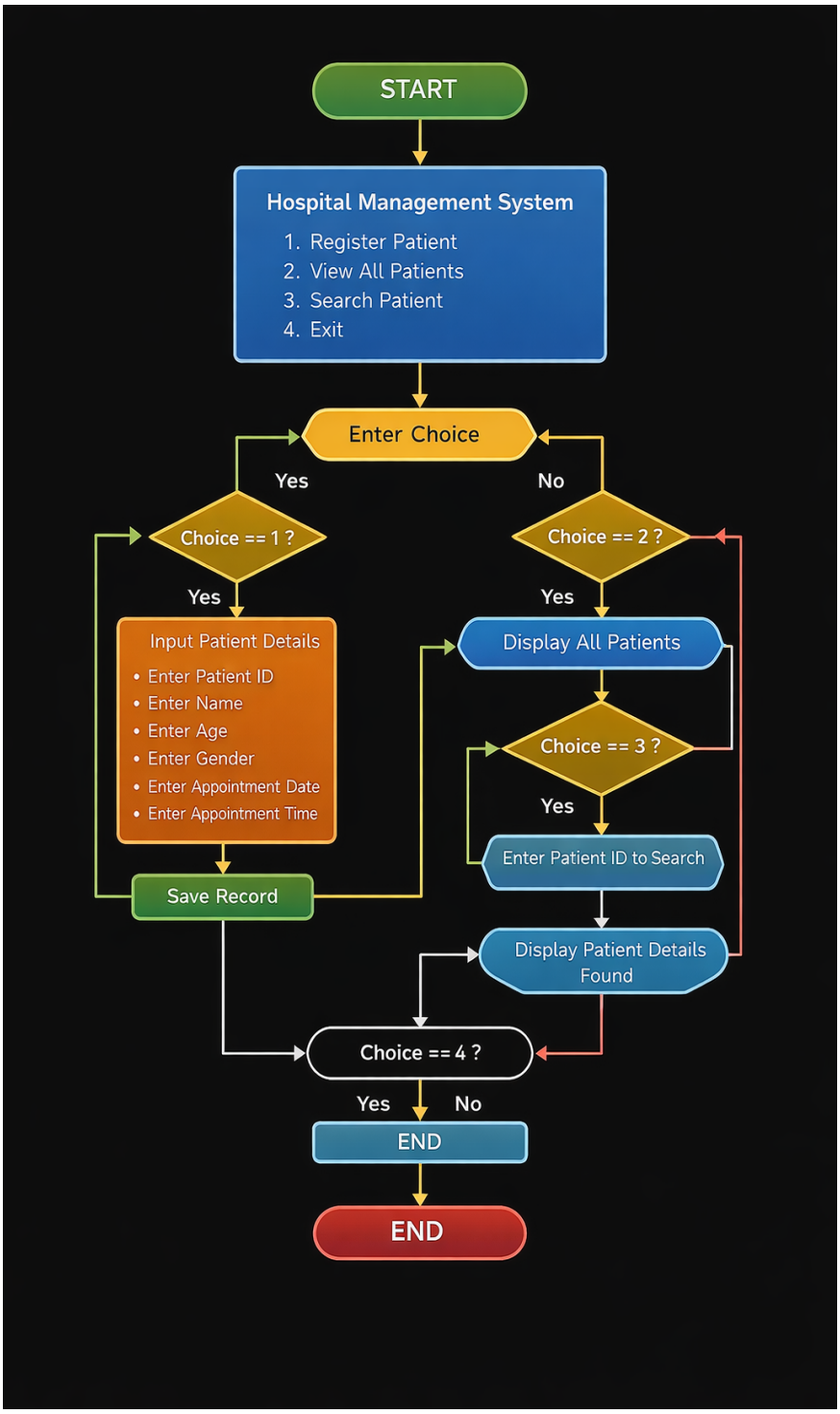
4.5 RESULT AND DOCUMENTATION

- System works efficiently
- Successfully manages patient data

4.6 ALGORITHM

1. Start
2. Display menu
3. Select option
4. Enter patient details
5. Store in file
6. Schedule appointment
7. Display details
8. Stop

4.7. Flowchart



5. SOURCE CODE

```
import java.io.*;
import java.util.*;

class Patient {
    int id;
    String name;
    int age;
    String gender;
    String date;
    String time;

    public Patient(int id, String name, int age, String gender, String date, String time) {
        this.id = id;
        this.name = name;
        this.age = age;
        this.gender = gender;
        this.date = date;
        this.time = time;
    }
}

public class HospitalManagement {

    static Scanner sc = new Scanner(System.in);
    static final String FILE_NAME = "patients.txt";
```

```
// Add Patient
static void addPatient() {
    try (FileWriter fw = new FileWriter(FILE_NAME, true);
         BufferedWriter bw = new BufferedWriter(fw)) {

        System.out.print("\nEnter Patient ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Age: ");
        int age = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Gender: ");
        String gender = sc.nextLine();

        System.out.print("Enter Appointment Date: ");
        String date = sc.nextLine();

        System.out.print("Enter Appointment Time: ");
        String time = sc.nextLine();

        bw.write(id + " " + name + " " + age + " " + gender + " " + date + " " + time);
        bw.newLine();
```

```
            System.out.println("\nPatient Registered Successfully!");

        } catch (IOException e) {
            System.out.println("Error writing to file!");
        }
    }

    // View Patients
    static void viewPatients() {
        File file = new File(FILE_NAME);

        if (!file.exists()) {
            System.out.println("\nNo records found!");
            return;
        }

        System.out.println("\n--- Patient Records ---");

        try (Scanner fileScanner = new Scanner(file)) {

            while (fileScanner.hasNextLine()) {
                String line = fileScanner.nextLine();
                String[] data = line.split(" ");

                if (data.length >= 6) {
                    System.out.println("\nID: " + data[0]);
                    System.out.println("Name: " + data[1]);
                    System.out.println("Age: " + data[2]);
                    System.out.println("Gender: " + data[3]);
```

```
                System.out.println("Date: " + data[4]);
                System.out.println("Time: " + data[5]);
                System.out.println("-------------------------");
            }
        }

    } catch (Exception e) {
        System.out.println("Error reading file!");
    }
}

// Search Patient
static void searchPatient() {
    File file = new File(FILE_NAME);

    if (!file.exists()) {
        System.out.println("\nNo records found!");
        return;
    }

    System.out.print("\nEnter Patient ID to search: ");
    int searchId = sc.nextInt();

    boolean found = false;

    try (Scanner fileScanner = new Scanner(file)) {

        while (fileScanner.hasNextLine()) {
            String line = fileScanner.nextLine();
```

```
            String[] data = line.split(" ");

            if (data.length >= 6 && Integer.parseInt(data[0]) == searchId) {
                System.out.println("\nRecord Found:");
                System.out.println("Name: " + data[1]);
                System.out.println("Age: " + data[2]);
                System.out.println("Gender: " + data[3]);
                System.out.println("Date: " + data[4]);
                System.out.println("Time: " + data[5]);
                found = true;
                break;
            }
        }

    } catch (Exception e) {
        System.out.println("Error reading file!");
    }

    if (!found) {
        System.out.println("\nRecord Not Found!");
    }
}

// Main Menu
public static void main(String[] args) {

    while (true) {
        System.out.println("\n--- Hospital Management System ---");
        System.out.println("1. Register Patient");
```

```
System.out.println("2. View All Patients");
System.out.println("3. Search Patient");
System.out.println("4. Exit");
System.out.print("Enter your choice: ");

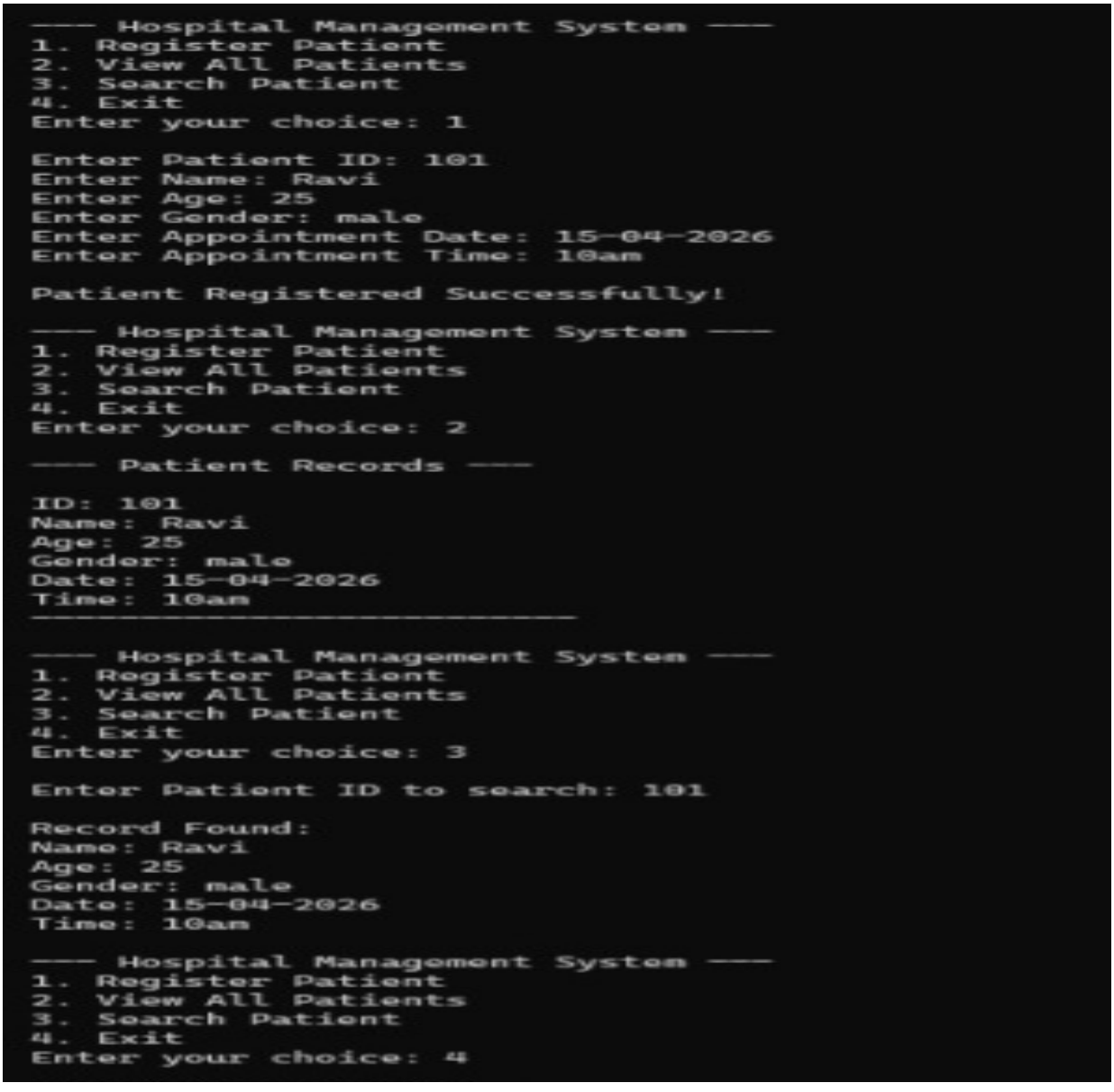
int choice;

try {
    choice = sc.nextInt();
} catch (InputMismatchException e) {
    System.out.println("Invalid input!");
    sc.nextLine();
    continue;
}

switch (choice) {
    case 1:
        addPatient();
        break;
    case 2:
        viewPatients();
        break;
    case 3:
        searchPatient();
        break;
    case 4:
        System.exit(0);
    default:
        System.out.println("Invalid choice!");
```

```
            }
        }
    }
}
```

6. INPUT AND OUTPUT SCREENS



7. DISCUSSION & CONCLUSION

7.1 DISCUSSION

The system simplifies hospital operations by automating registration and appointment scheduling. It reduces workload and improves efficiency.

7.2 CONCLUSION

The project successfully achieves its objectives by:

Reducing manual work

Improving accuracy

Saving time

Future Enhancements

Add graphical user interface (GUI)

Integrate database system

Enable online appointment booking

8. REFERENCES

Books

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Head First Java – Kathy Sierra and Bert Bates

Websites

https://www.geeksforgeeks.org

https://www.w3schools.com

https://www.javatpoint.com